

Digital Development in Africa

Internet Adoption, Mobile Connectivity and Infrastructure Across 11 Selected African Countries | 2010-2024

1. Objective / Problem Statement

This capstone project applies the complete data analytics workflow to examine digital development across 11 selected African countries. The project focuses on internet adoption, mobile connectivity, fixed broadband access, electricity access and secure internet infrastructure. The central problem is that digital access is not evenly distributed across African countries, and decision makers need evidence to understand where progress has occurred, where gaps remain, and which infrastructure areas require attention.

2. Dataset Description

The analysis uses the World Bank World Development Indicators (WDI) dataset. WDI is a broad global development dataset containing indicators across themes such as economy, education, health, environment, infrastructure, technology, governance and trade. The original dataset contained country identifiers, indicator names, indicator codes and yearly columns from 1960 to 2025. For this project, the dataset was narrowed to a digital-development theme and filtered to five indicators relevant to the analysis.

Selected Indicators

Indicator	Code	Purpose
Individuals using the Internet (% of population)	IT.NET.USER.ZS	Measures internet adoption
Mobile cellular subscriptions (per 100 people)	IT.CEL.SETS.P2	Measures mobile connectivity
Fixed broadband subscriptions (per 100 people)	IT.NET.BBND.P2	Measures fixed digital infrastructure
Access to electricity (% of population)	EG.ELC.ACCS.ZS	Measures enabling infrastructure
Secure Internet servers (per 1 million people)	IT.NET.SECR.P6	Measures digital infrastructure capacity

3. Data Cleaning Process

The raw WDI file was reduced to the selected theme by filtering to the five relevant indicator codes and 11 selected African countries. The year columns were unpivoted so that the dataset moved from wide format to long format. After unpivoting, each row represented one country, one indicator and one year. The unpivoted columns were renamed to Year and Indicator Value. Data types were corrected so that country and indicator fields were stored as text, Year was stored as a whole number and Indicator Value was stored as a decimal number. Missing values were not replaced with zero because a missing value means that no observation is available, while zero would represent an actual measured value. The main analysis period was set to 2010-2024 because all selected indicators have usable data within this period, and 2024 was used as the latest comparison year.

4. Analysis Methodology

The analysis followed a structured workflow: theme selection, indicator selection, country selection, data cleaning, data investigation, KPI creation, trend analysis, relationship analysis, dashboard design and business interpretation. The Power BI dashboard was designed as a one-page executive report showing KPI cards, trend analysis, country rankings, infrastructure relationships and key takeaways. The analysis focused on answering whether internet adoption has improved, which countries are ahead or behind, how mobile connectivity compares with actual internet usage, and whether infrastructure indicators such as electricity access and secure servers help explain remaining digital-development gaps.

5. Key Findings

The analysis indicates that digital access has expanded across the selected African countries between 2010 and 2024. Internet adoption improved substantially over the period, but the pace and level of progress differed across countries. Mobile cellular subscriptions are generally higher than internet usage, suggesting that mobile connectivity alone does not automatically translate into full digital participation. Fixed broadband remains much lower than mobile connectivity, showing that many countries still rely heavily on mobile-based access. Electricity access appears to be an important enabling infrastructure for digital adoption, while secure internet server availability remains uneven across countries.

6. Insights

The findings suggest that digital development is not only about whether people own mobile subscriptions. A country may have high mobile penetration while still having lower internet usage, limited fixed broadband and weaker secure internet infrastructure. This means that digital inclusion depends on a combination of connectivity, affordability, electricity, infrastructure quality and policy support. Countries with stronger electricity access and broader digital infrastructure are generally better positioned to expand meaningful internet adoption. The digital gap is therefore both an access problem and an infrastructure-readiness problem.

7. Recommendations

Stakeholders should prioritize expanding affordable broadband infrastructure, particularly in countries where mobile subscriptions are high but internet usage remains lower. Governments and development partners should continue investing in electricity access because reliable power supports digital participation. Countries with lower internet adoption should strengthen digital inclusion policies, affordability programs and rural connectivity initiatives. Mobile connectivity should be treated as an entry point, not the final measure of digital development. Secure internet infrastructure should also be improved to support digital services, online business, e-government and broader participation in the digital economy.

8. Conclusion

This project demonstrates how the complete analytics workflow can be applied to a large real-world development dataset. By narrowing the WDI dataset to a clear theme, selecting relevant indicators, transforming the data into a Power BI-friendly structure and developing a focused dashboard, the project provides evidence on Africa's digital-development progress and remaining gaps. The analysis shows that digital adoption is increasing, but infrastructure and access differences remain important barriers. The dashboard and recommendations can support policy, investment and development planning decisions.